

# Finishing I/O, and advanced I/O

Your program never talks to the disk, the network, or another program. It talks to the layer next to it. Every layer says *done* about something different [V50].

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# Roadmap — ten parts, one sequence

|            |                                |                                                                                      |
|------------|--------------------------------|--------------------------------------------------------------------------------------|
| <b>1</b>   | <b>Recap, on paper</b>         | You draw last week's stack.                                                          |
| <b>2–5</b> | <b>Lecture 2, finished</b>     | Kill demo · three levels · two kernels · buffered errors.                            |
| <b>6</b>   | <b>So when is it safe?</b>     | Last week's open question, measured.                                                 |
| <b>7–9</b> | <b>The announced tail</b>      | <a href="#">Waiting</a> · <a href="#">destinations</a> · <a href="#">zero-copy</a> . |
| <b>10</b>  | <b>Homework, next question</b> | What you can run this week.                                                          |

Parts 7, 8 and 9 are built in full. We may not reach them.

# Whose *done* did you measure?

One question, asked three times today. Take it with you.

## The rule

Almost every fast interface returns when it has accepted responsibility, not when the work happened [V50].

## Part 6

Three calls, three different places your bytes are. None is the platter.

## Part 7

Readiness says "you may read now". Completion says "I have read for you".

## Part 8

`write(3, ...)` says done when the bytes are in the pipe.

It works outside operating systems too. An HTTP 202 and a `git push` return before the work is done.

# Takeaways

- 1 Acknowledgement is not durability**  
`write()` reached a buffer. `fsync()` asked the device to flush. Neither proves the bytes are on the medium [V44].
- 2 Readiness and completion**  
One says you may act. The other acts for you [V77].
- 3 Three destinations**  
A file, a pipe and a socket are one file descriptor [V64] [V29].
- 4 One call, many buffers**  
`writew` crosses once with three buffers. `sendfile` moves bytes you never hold [S56].

Claims 2 to 4 are in the announced tail.

# What today does not cover

I measure the left column today. The right has its own lecture.

## TODAY

- 1 When file data reaches the device
- 2 What `fsync()` promises
- 3 Three ways to wait, counted
- 4 Files, pipes and sockets, traced
- 5 `writew` and `sendfile`, counted

## NOT TODAY

- 6 Writing your own `epoll` or completion-port loop
- 7 Memory mapping. Lecture 4.
- 8 Direct I/O, and ring programming
- 9 Crash dumps

# Why you write your answer down first

Watching a demonstration teaches no more than skipping it. Predicting first does [S51].

Crouch, Fagen, Callan and Mazur measured it in 2004: "students who passively observe demonstrations understand the underlying concepts no better than students who do not see the demonstration at all" [S51].

**no better**

watching, against a no-  
demonstration control [S51]

**predict  
first**

"significantly greater understanding"  
[S51]

One study, one subject. So each demonstration has three steps. You predict. We observe. We explain [S14].

PART 1

# Recap by prediction

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Three minutes, on paper, before I show you anything.

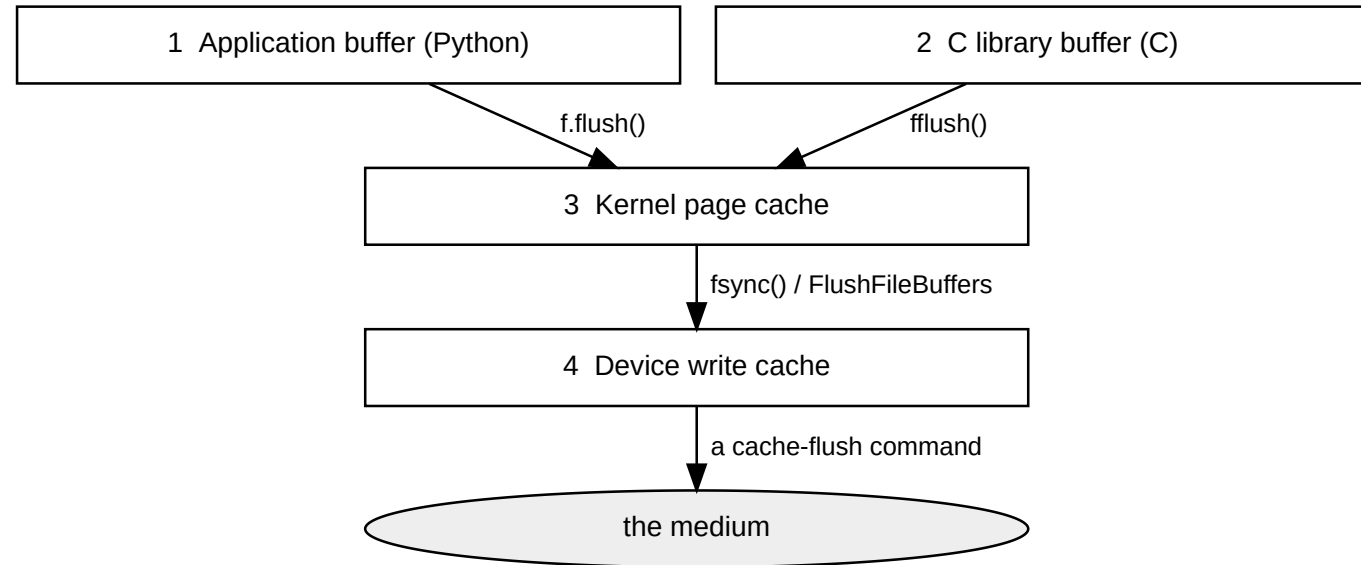
# Predict: draw last week's stack

Three minutes. Nothing on the screen until you have written something down.

- 1**                      **Draw it**                      The four levels between your program and the medium. Name each one.
- 2**                      **Mark it**                      After `fflush()` returns, which level are the bytes in?
- 3**                      **Answer it**                      The machine loses power right now. What is lost?  
Write all three. Nothing is handed in. I will ask several of you to read yours out [S14].

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/>

# It is not one buffer — it is a stack



Levels 1 and 2 are alternative routes into level 3 [V6]. Each arrow is one call [S17] [S3] [S13] [V37] [V33].

[S17] `io.IOBase.flush()` — Python documentation. [S3] `fflush(3)` — Linux man-pages. [S13] `fsync` — The Open Group.

# Explain: what dies with the process, what dies with the machine

Two different failures. Part 2's demonstration tests one. Your recap question asked about the other.

## THE PROCESS DIES

Levels 1 and 2 go with it.  
Anything already handed to the kernel survives, because the kernel writes it back on its own [V14].

## THE MACHINE LOSES POWER

Levels 1, 2 and 3 are gone. Level 4 goes too, unless the device has power-loss protection.

My own note is blunt: "the honest classroom demo is a power-cut, not a kill -9" [V14]. Part 2 opens with a kill -9. Hold the distinction. [Part 6 measures the level-3 window.](#)

# Predict now, and keep it for an hour

The prediction for Part 6. You write it now, before Part 2, so you carry the question through the carried hour.

```
# demo/durability/save.py, the gist
with open("saved.bin", "wb") as f:      # default buffering; no flush, no fsync
    for _ in range(100):
        f.write(record)                # one 64-byte record
print("done")                          # the program's done. Whose?
```

## The question

The process has exited. How long until the kernel actually writes your bytes to the device?

## Your options

Under a second · about five seconds · about thirty seconds · not until shutdown.

One of four. Write it in the corner of your page. Do not change it later [S14].

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/>

# What dies with the process

---

Parts 2 to 5 are Lecture 2, finished. Same slides, same order, renumbered into today.

## Where Lecture 2 stopped

Last Tuesday ended after slide 32: the device acknowledges early [V20], and no tool in the room can see whether it lied [V5].

From here to the end of Part 5 the slides are Lecture 2's, as built. The clock ran out after slide 32; slide 33, the `os._exit` demonstration, is where we pick up.

Your two predictions from Part 1 stay on your page: the level the bytes are in after `fflush()`, and what a power cut takes. The first demonstration answers a different question — what a *process* dying takes — so keep both apart.

# Three lines that print nothing

A demonstration. You have the stack now, so predict this one.

```
import os  
print("hello")  
os._exit(0)
```

**To a terminal**

Does hello appear?

**Redirected to a file**

What does the file contain?

Write down both answers before I run either one [S14].

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/>

# Why the file is empty

The rule sits in the C standard, not in the kernel.

- 1 Mode follows destination**

Output is "fully buffered if and only if the stream can be determined not to refer to an interactive device" [S5].
- 2 A terminal is interactive**

Bytes go out per line.
- 3 A file is not**

Bytes wait.
- 4 The exit skips cleanup**

No write reaches the kernel.

[S5] C99 draft §7.19.3. <https://busybox.net/~landley/c99-draft.html>

# What this demonstration does not prove

It proves level one is real. It proves nothing about levels three and four.

"the honest classroom demo is a power-cut, not a kill -9. Killing the process loses only the *runtime's* buffer; the kernel still writes back." [V14]

- |          |                         |                                                                   |
|----------|-------------------------|-------------------------------------------------------------------|
| <b>1</b> | <b>What died</b>        | The process, and the level-one buffer with it.                    |
| <b>2</b> | <b>What survived</b>    | Anything already in the page cache [V14].                         |
| <b>3</b> | <b>Wrong conclusion</b> | "So <b>flush() was enough.</b> " It was enough against this test. |

# Predict: what will `ltrace` show that `strace` did not?

Before I run the second tracer. Two lines on your page [S14].

It records the **library calls** a program makes [S31] — here, `fwrite` in the C library. `strace` sees `write` in the kernel instead.

- |          |                                                            |                                                                             |
|----------|------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>1</b> | <b>The C program calls <code>fwrite</code> three times</b> | How many <code>fwrite</code> lines does <code>ltrace</code> print?          |
| <b>2</b> | <b>The kernel saw one <code>write</code></b>               | Does <code>ltrace -S</code> print that line too, or only the library calls? |

The answer follows from the stack you drew twenty minutes ago.

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/> [S31] `ltrace(1)`. <https://man7.org/linux/man-pages/man1/ltrace.1.html>

# ltrace: the same idea, one layer up

strace stops at system calls. ltrace stops at calls into shared libraries.

## What it does

It "intercepts and records the dynamic library calls" a process makes [S31]. It can also print "the system calls executed by the program" [S31].

## Where it comes from

Juan Cespedes wrote it originally [S32]. The manual page names Petr Machata too [S31]. On Debian and Ubuntu you run `apt-get install ltrace` [S32].

## HOW TO USE IT

```
ltrace ./iodemo      # the C library calls: fopen, fwrite, fclose
ltrace -S ./iodemo   # "system calls as well as library calls" [S31]
ltrace -c ./iodemo   # a count for each library call [S31]
```

It runs on Linux only.

[S31] ltrace(1). <https://man7.org/linux/man-pages/man1/ltrace.1.html> [S32] ltrace. <https://ltrace.org/>

# Which levels can you actually watch?

I promised three of four boundaries. Here is the tool each needs.

| BOUNDARY         | WHAT WATCHES IT                                            | IN THIS COURSE |
|------------------|------------------------------------------------------------|----------------|
| App → library    | ltrace sees it, strace does not [V10]                      | the tool above |
| Library → kernel | strace — where our tracer attaches [S1]                    | today          |
| Kernel → device  | a tracer for the kernel's block layer [V30]                | elsewhere      |
| Device → medium  | no system-call trace shows whether the device flushed [V5] | no tracer      |

# The same write, at three levels

---

Now that the levels have a reason to exist, walk down them with one small program.

# The same write, at three levels

| LEVEL       | PYTHON                           | C                             | BUFFERED? |
|-------------|----------------------------------|-------------------------------|-----------|
| Application | <code>f.write(b"x")</code> [S6]  | —                             | yes [S6]  |
| C library   | —                                | <code>fwrite(...)</code> [S3] | yes [S3]  |
| System call | <code>os.write(...)</code> [S20] | <code>write(...)</code> [S2]  | no        |

The bottom row takes a **file descriptor** — the small integer naming an open file. You saw one already: the `= 3` in the trace line was a file descriptor [S1].

[S1] `strace(1)` — Linux man-pages. <https://man7.org/linux/man-pages/man1/strace.1.html> [S2] `write(2)` — Linux man-pages. <https://man7.org/linux/man-pages/man2/write.2.html> [S3] `fflush(3)` — Linux man-pages. [S6] Python documentation, `open()`. [S20] `os` — Miscellaneous operating system interfaces, `os.write()`. <https://docs.python.org/3/library/os.html#os.write>

# Meet the program: `iodemo.c`

Nine lines of C. The first open fails on purpose, so the trace shows an error path.

```
f = fopen("no-such-file.txt", "rb"); /* 1. deliberate failure */
if (f != NULL) { fclose(f); return 2; }

f = fopen(path, "wb");                /* 2. the real file */
if (f == NULL) { return 1; }

fwrite("one\n", 1, 4, f);              /* 3. three writes */
fwrite("two\n", 1, 4, f);              /*    fourteen bytes */
fwrite("three\n", 1, 6, f);

if (fclose(f) != 0) { return 3; }      /* 4. flush, then close */
```

It prints nothing, on purpose. A program that prints nothing adds no write lines of its own to the trace.

# Three calls in the source. One in the trace.

# 3

`fwrite` calls in the source

# 14

bytes written in total

# 1

`write` call in the trace

The three `fwrite` calls go into one C library buffer. The manual calls that level "the user-space buffers provided by the C library" [S3]. The line to look for in the live trace, predicted from the source:

```
write(3, "one\ntwo\nthree\n", 14) = 14
```

The 3 is the file descriptor. The 14 is all three `fwrite` calls, in one crossing.

[S3] `fflush(3)` — Linux man-pages. <https://man7.org/linux/man-pages/man3/fflush.3.html>

# fwrite takes a size and a count

Four arguments, and two of them multiply.

```
fwrite("three\n", 1, 6, f);  
/*      data      ^  ^  stream  
               |  +-- number of items  
               +----- size of one item, in bytes */
```

## 1 It returns a count

Items written, not bytes. The count equals bytes only when the item size is 1 [S21].

## 2 Fewer is legal

Below it, "a successful write() may transfer fewer than count bytes" [S2].

[S2] write(2) — Linux man-pages. <https://man7.org/linux/man-pages/man2/write.2.html> [S21] fread(3), which documents fwrite(3) — Linux man-pages. <https://man7.org/linux/man-pages/man3/fwrite.3.html>

# Two operating systems, side by side

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Read as a comparison. Try the tools yourself afterwards.

# Process Monitor: what the kernel did

It reports filesystem and registry activity [S18], which is a different cut than the system-call boundary.

## What it does

It shows "real-time file system, Registry and process/thread activity" [S18]. That is a filesystem-and-registry view, not a list of system calls.

## Where it comes from

Mark Russinovich wrote the page, in Microsoft's Sysinternals documentation [S18]. The tool "combines the features of two legacy Sysinternals utilities, Filemon and Regmon" [S18]. You download it. You can also run it from Sysinternals Live [S18].

## HOW TO USE IT

```
Procmon.exe          # it starts capturing at once
Filter → Process Name is iodemo.exe # keep one process
Operation column     # expect CreateFile and WriteFile rows
```

[S18] Process Monitor. <https://learn.microsoft.com/en-us/sysinternals/downloads/procmon>

# NtTrace: a tracer for the Windows Native API

It watches the native API, "the interface between the application space and the OS kernel" [S10].

## What it does

A "simple trace facility for the Windows Native API", "roughly equivalent to strace on Linux" [S10]. It attaches as a debugger, so it "needs no driver and no administrator rights" [V21].

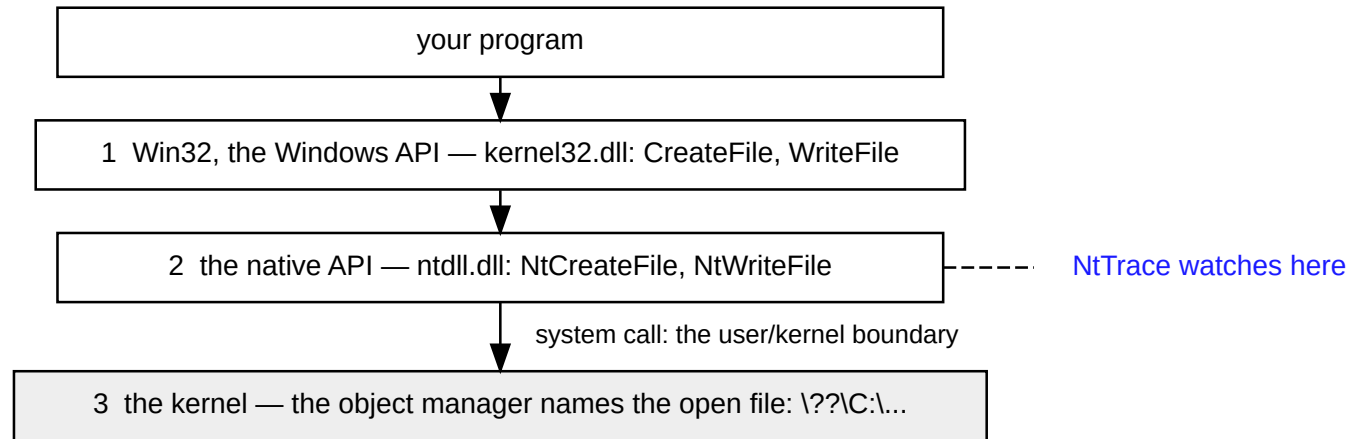
## HOW TO USE IT

```
NtTrace iodemo.exe           # one line per native call
NtTrace -filter File iodemo.exe # keep calls whose names contain "File" [S10]
NtTrace -only iodemo.exe      # not its child processes [S10]
```

Try both yourself on `iodemo.exe`: Process Monitor shows what the filesystem did, NtTrace shows what the program asked for.

[S10] NtTrace — Roger Orr. <https://rogerorr.github.io/NtTrace/> [S33] NtTrace source. <https://github.com/rogerorr/NtTrace>

# Three Windows words you will need



- 1 Win32** The documented interface a Windows program calls. The 32 is history, not a width [S22].
- 2 The native API** The layer below Win32, inside `ntdll.dll`. Microsoft calls it "internal to the operating system" [S9].
- 3 The object manager** The kernel component that names open objects. Its paths begin `\\??\\` [V22].

[S9] `winternl.h` — Microsoft Learn. <https://learn.microsoft.com/en-us/windows/win32/api/winternl/nf-winternl-ntquerysysteminformation> [S22] Windows API index — Microsoft Learn. <https://learn.microsoft.com/en-us/windows/win32/apiindex/windows-api-list>

# Predict: the same program, on Windows

Before the Windows capture goes on the screen. Two answers, on paper [S14].

## 1 The three fwrite calls

How many write calls does the Windows trace show — three, or one?

## 2 The deliberate failure

Linux reported -1 ENOENT. How many error codes does Windows report for the same failed open?

The first answer tests whether buffering is a C library property or a platform property. The second one has a surprise in it.

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/>

# A real Windows trace

You can already read this. The anatomy matches the Linux line, and the vocabulary differs.

```
NtCreateFile(... ObjectAttributes=0x58:"no-such-file.txt", ...)
    ⇒ 0xc0000034 [2 'The system cannot find the file specified.']
NtCreateFile(FileHandle=... [0x124], ... ObjectAttributes=0x58:"iodemo.out", ...) ⇒ 0
NtWriteFile(FileHandle=0x124, ..., Length=0xe, ...) ⇒ 0
NtClose(Handle=0x124) ⇒ 0
```

`iodemo.exe` under NtTrace, 2026-09-15 — the same four steps as the Linux trace [S36] [S37]. Ignore every argument except the handle, the file name and the return value.

[S36] Our NtTrace64 capture of `iodemo.exe`, 2026-09-15 — `demo/iodemo-c/NtTrace64-IODEMO-Output.txt` [S37] The container capture of `iodemo`, 2026-09-15 — `demo/iodemo-c/README.md`

# A handle is not a file descriptor

First difference, visible in the first argument of almost every line.

## LINUX

A file descriptor. A small integer indexing a per-process table. Our trace shows = 3 [S37].

## WINDOWS

A handle. An opaque value naming an open object. Our trace shows  
FileHandle=0x124 [S36].

Same job in a trace, different mechanism in the system. Do not use one word for the other.

[S36] Our NtTrace64 capture of iodemo.exe, 2026-09-15 — demo/iodemo-c/NtTrace64-IODEMO-Output.txt [S37] The container capture of iodemo, 2026-09-15 — demo/iodemo-c/README.md

# Two error namespaces in one return value

Second difference, and it is the layer boundary showing up where nobody expects it.

```
⇒ 0xc0000034 [2 'The system cannot find the file specified.']
```

- |          |                      |                                             |
|----------|----------------------|---------------------------------------------|
| <b>1</b> | <b>First number</b>  | An NTSTATUS, the native layer's code [S36]. |
| <b>2</b> | <b>Second number</b> | The Win32 error it translates into [S36].   |
| <b>3</b> | <b>Linux</b>         | One errno: -1 ENOENT [S37].                 |

[S36] Our NtTrace64 capture of iodemo.exe, 2026-09-15 — demo/iodemo-c/NtTrace64-IODEMO-Output.txt [S37] The container capture of iodemo, 2026-09-15 — demo/iodemo-c/README.md

# The stable boundary sits at a different level

Third difference, and the other two are symptoms of it.

In our trace of the Windows command shell a path reads "\\??\\C:\\WINDOWS\\system32\\c\_852.nls" [S34]. That prefix "is the object-manager namespace, which a Win32 program never types" [V22].

## LINUX

[V8]

The system  
call interface  
is the public  
contract.

Syscall

## WINDOWS

[S9]

Native entry  
points are  
"subject to  
change from  
one release of  
Windows to  
another".

Win32

[S9] winternl.h — Microsoft Learn. <https://learn.microsoft.com/en-us/windows/win32/api/winternl/nf-winternl-ntquerysysteminformation>

# Where the boundary sits is contested history

Two readings survive the evidence. I will not pick one for you.

- 1**                      **Reading A: deliberate**                      POSIX, OS/2 and Win32 each reached NT through the native API [S19].
- 2**                      **Reading B: vestige**                      A reported POSIX subsystem that was "never really completed" [S15].
- 3**                      **Linux had no choice**                      It "had to make the kernel boundary itself the stable interface" [V38].

[S15] The sad history of the Microsoft POSIX subsystem — Reiter (2010). <https://brianreiter.org/2010/08/24/the-sad-history-of-the-microsoft-posix-subsystem/> [S19] Inside Native Applications — Russinovich (2006). <https://learn.microsoft.com/en-us/sysinternals/resources/inside-native-applications>

# Linux is the outlier, not Windows

Only Linux promises that the raw system call interface stays the same [V9]. Say *different*, never *worse*.

- 1 Who makes no such promise** FreeBSD, OpenBSD, NetBSD, Solaris/illumos, macOS [V43].
- 2 Not promised is not broken** FreeBSD keeps old programs running through compatibility layers; it never writes the promise down [V36].
- 3 Why Windows and macOS can** They ship their own C library, so the library is the stable surface and the kernel boundary can move [V38].

# One program, two kernels

One program, two systems, traced on one day.

| STEP        | LINUX [S37]                                                 | WINDOWS [S36]                                                                |
|-------------|-------------------------------------------------------------|------------------------------------------------------------------------------|
| failed open | <code>openat(... "no-such-file.txt" ...) = -1 ENOENT</code> | <code>NtCreateFile(... "no-such-file.txt" ...) ⇒ 0xc0000034 [2 '...']</code> |
| open        | <code>openat(... "iodemo.out" ...) = 3</code>               | <code>NtCreateFile(... [0x124] ... "iodemo.out" ...) ⇒ 0</code>              |
| write       | <code>write(3, "one\ntwo\nthree\n", 14) = 14</code>         | <code>NtWriteFile(FileHandle=0x124, ... Length=0xe ...) ⇒ 0</code>           |
| close       | <code>close(3) = 0</code>                                   | <code>NtClose(Handle=0x124) ⇒ 0</code>                                       |

Same program, same fourteen bytes, **one write on each side**. Hold the program fixed, and what is left is the two systems.

[S36] Our NtTrace64 capture of `iodemo.exe`, 2026-09-15 — `demo/iodemo-c/NtTrace64-IODEMO-Output.txt` [S37] The container capture of `iodemo`, 2026-09-15 — `demo/iodemo-c/README.md`

# What we have not looked at

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One question I will not answer, and the work that follows from it.

# **write() returned success. The bytes are not on a disk.**

I built the whole hour towards this sentence. I leave it unresolved.

The manual says it plainly: "A successful return from write() does not make any guarantee that data has been committed to disk" [S2].

Saltzer, Reed and Clark gave the general form in 1984. A lower level may offer "an incomplete version of the function... useful as a performance enhancement" [S11].

**So when is it actually safe?** That is the durability lecture's question. I leave it open today.

[S2] write(2) — Linux man-pages. [S11] End-to-End Arguments in System Design — Saltzer, Reed and Clark (1984).  
<https://web.mit.edu/Saltzer/www/publications/endtoend/endtoend.pdf>

# Buffering moves your errors

The error arrives somewhere other than where you caused it.

- |   |                           |                                                                                                               |
|---|---------------------------|---------------------------------------------------------------------------------------------------------------|
| 1 | <b>Why it moves</b>       | The bytes only reached a buffer.                                                                              |
| 2 | <b>Where it arrives</b>   | Where the library flushes, so at <code>fclose</code> .                                                        |
| 3 | <b>The rule</b>           | Check the return value of the call that flushes.                                                              |
| 4 | <b>The idiom hides it</b> | "the most common durability idiom in every mainstream language... contains no durability step anywhere" [V4]. |

# Predict: all your writes succeeded — can the close still fail?

Yes or no, on paper. Then a second question: if it fails, is the descriptor still yours?

- |          |                                                           |                                                                  |
|----------|-----------------------------------------------------------|------------------------------------------------------------------|
| <b>1</b> | <b>Every <code>write()</code> returned the byte count</b> | The program did nothing wrong that it could see.                 |
| <b>2</b> | <b>Now <code>close()</code></b>                           | Can it return -1? On what kind of filesystem?                    |
| <b>3</b> | <b>If it did</b>                                          | Is the file descriptor number still open, closed, or unknowable? |

Two answers, written down [S14].

# Why a close can fail when every write succeeded

The kernel deferred the write's error. The close surfaces it [S4].

- 1 Deferred** "errors on a previous write(2) operation are reported only on the final close()" [S4].
- 2 Where it bites** "silent loss of data ... with NFS and with disk quota" [S4].
- 3 The mirror** "A successful close does not guarantee that the data has been successfully saved to disk" [S4]. Part 6's question.

Against the stack: `write()` said *done* at level 3 [S2]. `close()` can still refuse [S4].

[S4] `close(2)` — Linux man-pages. <https://man7.org/linux/man-pages/man2/close.2.html>

# Observe: close it twice, and watch what the second close hits

demo/close/ — one descriptor, two closes, both platforms [S39].

```
Linux, strace      close(3) = 0, then close(3) = -1 EBADF (Bad file descriptor)
Windows, untraced first CloseHandle ok, then error 6 (ERROR_INVALID_HANDLE)
```

The *first* close released the number, error or not: Linux "always releases the file descriptor early in the close operation, freeing it for reuse" [S4]. So a retry closes nothing — or, with threads, "a reused file descriptor from another thread" [S4].

Under a debugger `CloseHandle` "will throw an exception if it receives ... a handle value that is not valid ... This can happen if you close a handle twice" [S38]. *NtTrace* is a debugger, so the second close raises [S39].

[S4] `close(2)`. <https://man7.org/linux/man-pages/man2/close.2.html> · [S38] `CloseHandle`. <https://learn.microsoft.com/en-us/windows/win32/api/handleapi/nf-handleapi-closehandle> · [S39] demo/close/README.md. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/close/README.md>

# Never retry a failed close

The obvious repair is wrong, and the manual page says so [S4].

"Retrying the close() after a failure return is the wrong thing to do, since this may cause a reused file descriptor from another thread to be closed" [S4].

A retry closes whatever now sits at that number, which may be another thread's file [S4].

[Report the error and stop](#). Part 6 shows the same rule one level down, at fsync, and the data loss that taught it [V40].

[S4] close(2) — Linux man-pages. <https://man7.org/linux/man-pages/man2/close.2.html>

# Networking and IPC are this same mechanism

Not new subjects. The same objects and the same calls, wearing different names.

**1**                      **A socket on Linux**                      It is a file descriptor, and Linux makes that contract binding [V28].

**2**                      **A socket on Windows**                      An opaque SOCKET, and Windows makes Winsock the contract [V28].

**3**                      **A pipe**                      A file descriptor too, so the same calls reach it [V29].

Depth belongs to the next lecture. I want to stop you filing networking under "something else".

# Honest limits

What today's tools cannot see.

- **The device level** [V5]  
Not observable from software: no trace can check it.
- **Asynchronous I/O** [V11]  
The tracer sees the setup, then no I/O.

# Honest limits, continued

And what today's tools cost.

- **strace in production**

[V25]

Even the cheaper `--seccomp-bpf` mode [V39] "should therefore not be used on production systems!"

- **Mapping, crash dumps**

Each has its own lecture.

# Lecture 2 ends here. The question does not.

`write()` returned success and the bytes are not on a disk [S2] — so when is it safe? Part 6 measures it.

Part 5 named the question and the honest limits of the tools. Your prediction for Part 6 has been on your page since Part 1: under a second, about five, about thirty, or not until shutdown.

# So when is it safe?

---

The question I left you with last week. Your prediction is already on your page.

# The kernel's own count of what it has not written

No tracer here. The kernel publishes the number itself, and we watch it fall.

```
make container-watch          # demo/durability/  
# save.py writes 100 records, prints "done", exits  
# watch_dirty.py then samples /proc/meminfo every 0.5 s
```

## Dirty

File data the kernel holds and has not written to the device yet. A page in this state is a **dirty page** [V78].

## Writeback

Data the kernel is writing to the device right now [V78].

It is a machine-wide counter, so we watch it rise when `save.py` runs and fall when the kernel is finished [S52].

[S52] [demo/durability/README.md](https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md) — captured 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md>

# Observe: the program finished at zero seconds

```
vm.dirty_expire_centisecs    = 3000    (30 s)
vm.dirty_writeback_centisecs = 500     (5 s)

t= 0.0s Dirty=    1176 kB baseline
t= 0.0s Dirty=    5132 kB ← save.py has EXITED: "done: 100 records x 40960 B"
t= 0.5s Dirty=    5284 kB
t= 2.5s Dirty=    5364 kB
t= 4.0s Dirty=    5388 kB
t= 5.0s Dirty=     676 kB ← the kernel has finished
```

## 5.0 s

between the program's *done* and the kernel's, for a 4 MiB file [S52]

[S52] demo/durability/README.md — captured writeback timeline, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md>

# Five runs, three sizes

One run is an anecdote. Here are all five, including the one that disagrees.

| WRITTEN                                       | RUNS                                    | THE KERNEL'S <i>DONE</i> , AFTER THE PROGRAM'S |
|-----------------------------------------------|-----------------------------------------|------------------------------------------------|
| $100 \times 40 \text{ KiB} = 4 \text{ MiB}$   | 3 (two on overlayfs, one on plain ext4) | <b>5.0 s, 5.5 s, 5.5 s</b>                     |
| $100 \times 64 \text{ B} = 6 \text{ KiB}$     | 1                                       | <b>5.5 s</b>                                   |
| $1000 \times 40 \text{ KiB} = 40 \text{ MiB}$ | 1                                       | <b>0.5 s</b>                                   |

Four of the five runs land at five seconds, and file size barely matters between six kilobytes and four mebibytes [S52]. The forty-mebibyte run went out ten times sooner, which the next slide explains.

[S52] demo/durability/README.md — five runs, captured 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md>

# Five seconds in memory that belongs to nobody

The program's *done* and the kernel's *done* are different events, five seconds apart.

- 1**      **Where the bytes were**      In the **page cache**, level 3 of your drawing. That memory is the kernel's [V78].
- 2**      **What set the delay**      A kernel timer, not your program and not Python [V78].
- 3**      **What a power cut takes**      Everything in that window [V14].  
The return value was correct. It reported that the kernel had accepted the bytes [V50].

# Which rule fired, and how we know

The setting a student reads first is not the one that applied.

|                                     |                                                                                         |
|-------------------------------------|-----------------------------------------------------------------------------------------|
| <code>dirty_expire_centisecs</code> | This machine: 3000, so 30 s [S52]. It marks when data is "eligible for writeout" [S68]. |
|-------------------------------------|-----------------------------------------------------------------------------------------|

|                                        |                                                                                                    |
|----------------------------------------|----------------------------------------------------------------------------------------------------|
| <code>dirty_writeback_centisecs</code> | This machine: 500, so 5 s [S52]. It sets how often the "kernel flusher threads ... wake up" [S68]. |
|----------------------------------------|----------------------------------------------------------------------------------------------------|

|                       |                                                                              |
|-----------------------|------------------------------------------------------------------------------|
| <b>The 40 MiB run</b> | Above <code>dirty_background_ratio</code> , "writeback starts sooner" [V48]. |
|-----------------------|------------------------------------------------------------------------------|

**Writeback** is the kernel copying dirty pages out. Which rule fired here is my reading.

[S68] [admin-guide/sysctl/vm.html](https://www.kernel.org/doc/html/latest/admin-guide/sysctl/vm.html). <https://www.kernel.org/doc/html/latest/admin-guide/sysctl/vm.html> [S52] [demo/durability/README.md](https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md).  
<https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md>

# Visibility is not durability

The objection you are about to make, and the answer to it.

## VISIBILITY

After a successful `write()`, a later read returns the bytes. Another process reading the file sees them too. This is a coherence property of the page cache [V45].

## DURABILITY

The bytes survive a power loss. This is a property of the medium [V45].

My own note calls confusing the two "the single most common error in this area" [V45]. So `cat results.csv` returning your data proves visibility. It proves nothing about the disk.

# Predict: what does one `fsync` per record cost?

Before the counts. Same program, same hundred records, four ways [S14].

## Measured

One `f.flush()` per record. 101 write calls, **1.1 ms** of wall time [S52].

## Your prediction

Add one `os.fsync()` per record — the third arrow on the stack you drew, level 3 to level 4. How long?

Write a number in milliseconds. Not a factor, a number.

[S52] [demo/durability/README.md](https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md) — four-mode counts, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md>

# Observe: sixty times, for the same hundred writes

| MODE                    | WRITE CALLS | FSYNC CALLS | WALL TIME      |
|-------------------------|-------------|-------------|----------------|
| default                 | 6           | 0           | 0.2 ms         |
| flush per record        | 101         | 0           | 1.1 ms         |
| <b>fsync per record</b> | 101         | <b>100</b>  | <b>67.3 ms</b> |
| unbuffered              | 101         | 0           | 1.0 ms         |

Sixty times the wall time, for the same hundred write calls plus a hundred fsync calls [S52]. The cost is the waiting, not the extra crossings.

Microsoft concedes the same: FlushFileBuffers "can be inefficient when used after every write to a disk drive device" [S40]. Most programs never pay it.

[S52] demo/durability/README.md — four-mode counts, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md> [S40] FlushFileBuffers function (fileapi.h) — Microsoft Learn. <https://learn.microsoft.com/en-us/windows/win32/api/fileapi/nf-fileapi-flushfilebuffers>

# What `fsync()` promises, and what it does not

It crosses one boundary. The next is not yours to check.

- |   |                     |                                                                                              |
|---|---------------------|----------------------------------------------------------------------------------------------|
| 1 | <b>Does</b>         | Transfers the file's data "to the disk device", "including ... flushing a disk cache" [S67]. |
| 2 | <b>Waits for</b>    | It "blocks until the device reports that the transfer has completed" [S67].                  |
| 3 | <b>Cannot check</b> | Whether the device flushed. No trace tells "an honest flush from a fast lie" [V5].           |

Durability is a sequence of flushes, each crossing one boundary, each able to fail alone [V44]. [The device reports.](#)

[S67] `fsync(2)` — Linux man-pages. <https://man7.org/linux/man-pages/man2/fsync.2.html>

# Windows: the same shape, and now measured

The Windows twin of the four-mode run, on my Windows 11 VM [S52].

| IOCost.exe, 100 × 4 KiB            | KERNEL WRITE OPS | WALL           |
|------------------------------------|------------------|----------------|
| WriteFile into the cache           | 100              | 0.7 ms         |
| <b>FlushFileBuffers per record</b> | 100              | <b>27.9 ms</b> |
| FILE_FLAG_WRITE_THROUGH            | 100              | 25.2 ms        |
| FILE_FLAG_NO_BUFFERING             | 100              | 9.9 ms         |

[Forty times](#), for asking the device to confirm each record [S52]. Linux's figure was sixty. The two ratios are not comparable: these records are 4 KiB, the Linux ones were 64 B [S52].

Now read the count column: 100 in every row.

[S52] demo/durability/README.md, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md> [S40] FlushFileBuffers — Microsoft Learn. <https://learn.microsoft.com/en-us/windows/win32/api/fileapi/nf-fileapi-flushfilebuffers>

# The two kernels write in opposite orders

The same watch, on Windows, until the counter returned to baseline [S52].

| WRITTEN           | WINDOWS CACHE MANAGER | LINUX KERNEL          |
|-------------------|-----------------------|-----------------------|
| 4 MiB, three runs | 0.5 s · 1.0 s · 0.5 s | 5.0 s · 5.5 s · 5.5 s |
| 40 MiB, one run   | 14.1 s                | 0.5 s                 |

Windows' **lazy writer** had the small file out inside a second. Linux waited for its flusher's five-second wake. On the large file they swap — Linux wrote it at once, Windows trickled 512 pages a pass [S52].

Both clocks are a kernel's. Neither is your program's.

[S52] demo/durability/README.md, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md>

# Predict: your `fsync()` failed — what does the retry return?

The device rejected the write, so `fsync()` returned `-1` with `EIO`. The obvious repair: call it again.

**1**                      **The second call**                      `-1` again · 0, bytes on the device · 0, bytes gone.

**2**                      **Who found out**                      A database team, in 2018, in production [S41].

One of three, written down next to your Part 1 prediction [S14].

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/> · [S41] `fsyncgate`. <https://danluu.com/fsyncgate/>

# Observe: the retry returns success, over data that was never written

2018, and the case is known as *fsync-gate*. A PostgreSQL developer found that retrying a failed `fsync()` "is not OK at least on Linux" [S41].

```
fsync(3) = -1 EIO      ← the kernel reports the failure, once
close(3)
open(...)
fsync(3) = 0          ← success, over data that was never written
```

That is a real reproduction against an error-injecting device. Its author annotated the last line "fsync on file with still-dirty data+EIO returns success BAD", for a byte that "certainly never got written" [V53].

[S41] fsyncgate — the pgsql-hackers thread, archived. <https://danluu.com/fsyncgate/>

# Explain: the retry does not retry

Mechanical, not sloppy code: **the retry does not re-attempt the failed I/O** [V54].

1. **Reported once.** The kernel marks the page failed, reports it to the first `fsync()`, and clears the mark [S41].
2. **Already gone.** A failed writeback drops the data and marks the page clean, so a retry has nothing to write [V54].
3. **So the retry lies.** "It reads a status register that the previous read destroyed, over a page cache the data has already left" [V89].

The result: "how PostgreSQL lost data silently for twenty years" [V90].

# What the engines did instead

Three storage engines, three architectures, one answer, in one year.

|            |       |                                    |      |
|------------|-------|------------------------------------|------|
| POSTGRESQL | [V40] | Retry replaced by a crash [V93].   | 2018 |
| INNODB     | [V84] | Retry loop became a fatal error.   | 2018 |
| WIREDTIGER | [V24] | No retry. Panic on a failed flush. | 2018 |

A stopgap that "has now lasted eight years" [V56].

# Windows here is not "the same", and not "safe"

It is undocumented, which is a third answer and a worse one.

- **What is missing**

[V47]

The PostgreSQL wiki's matrix of post-fsync-failure behaviour has no Windows row at all.

- **Why nobody can settle it**

[V55]

The Linux contract was inspectable, so it was discoverable. The Windows contract "cannot be settled by reading" [V55].

Linux documents a hostile contract: one error report, then the data is gone. That is actionable. "For a system that must not lose data, an unspecified contract is the worse of the two positions" [V91].

# Sync the directory too

This part is yours, and the classic recipe is incomplete [V51].

A file is two pieces of data in two places: its **bytes**, and its **name**, which lives in the directory. `fsync()` on the file flushes the bytes. The manual: "Calling `fsync()` does not necessarily ensure that the entry in the directory containing the file has also reached disk" [S67].

So "create-write-fsync-close on a new file is the classic incomplete recipe" [V51]: after a crash the bytes are safe and the file may have no name.

The fix costs one line: open the directory, and `fsync()` that descriptor too [S67].

# Write-then-rename, and why it needs a sync

How every editor saves, and how ext4 made it lose files in 2009 [S44].

```
write a temporary file → close it → rename(temp, original)
```

**Why the pattern.** `rename()` swaps the name atomically: "there is no point at which another process attempting to access newpath will find it missing" [S70].

**The trap.** The rename is a small piece of metadata. The temporary file's bytes are still dirty in the page cache. ext4 wrote the rename first, so a crash meant "that existing file gets replaced with a zero-length file" [S44].

**The fix.** `fsync()` the temporary file *before* the rename [S44]. Then the name can only point at bytes already on the device.

# The respectable opposite view

Most programs are right not to pay, until the failure is "*silent*" [V52].

- 1 Most need not pay**

SQLite's own write-ahead-log setting "does lose durability", yet "a good choice for most applications" [S43].
- 2 Trusting the layer**

Trusting the layer below is "a defensible ... position", and "the implicit default" [V52].
- 3 Blame the API**

"no one ever, ever wrote `creat()`; `write()`; `close()`; `rename()`; and hoped they would get an empty file" [S45].

[S43] SQLite. [https://www.sqlite.org/pragma.html#pragma\\_synchronous](https://www.sqlite.org/pragma.html#pragma_synchronous) [S45] Aurora, LWN (2009). <https://lwn.net/Articles/351422/>

# Log first — and what if the log write fails?

The write-ahead log protects you because of an *order*, and the order covers its own failure.

1. **Order.** Log first, `fsync`d; only then *committed* [V85]. The data file follows later, at a checkpoint [S71].
2. **Log `fsync` fails:** nothing was promised yet. PostgreSQL panics here too: "we already did the same for WAL data" [S71].
3. **Data-file `fsync` fails later:** "the only remaining copy of the data is in the WAL" [S71]. Crash, replay.

The log wins only because it is made durable *first*, one `fsync` per commit — the 67 ms you saw [S52].

# So when is it safe?

The answer, and what it does not cover.

## In software

The last *done* you see is `fsync()`, resting on the device's report [S67].

## In hardware

A tested UPS keeps the host up. Level 4 is inside the drive, and needs its own power-loss protection.

## Neither covers

`fsync-gate` lost data **with the power on** [V53]. No battery helps.

## So also

Log every change first: "recover from the WAL after any failure" [V85].

SQLite trusts neither call — "neither ... works as advertised" [V49].

# Lecture 4 starts here

---

Parts 7, 8 and 9 are built in full. If the clock has gone, Lecture 4 opens on this slide and we jump to part 10.

# What a pipe is

The thing we are about to wait on, a hundred times over.

"Pipes and FIFOs (also known as named pipes) provide a unidirectional interprocess communication channel. A pipe has a *read end* and a *write end*" [S59].

- A pipe is a 64 KiB buffer inside the kernel [S59]. One process writes into one end, another reads from the other.
- Both ends are file descriptors, so `read()` and `write()` work on them unchanged [S59].
- "If a process attempts to read from an empty pipe, then `read(2)` will block until data is available" [S59]. That word, *block*, is the next slide.

# Blocking, non-blocking, and letting the kernel wait for you

Three ways to ask a pipe "is there anything to read?"

1. **Blocking.** `read()` on an empty pipe parks your thread until data arrives [S59]. Fine for one pipe, hopeless for a hundred.
2. **Non-blocking.** Set `O_NONBLOCK`; now `read()` returns at once, with `EAGAIN`, when there is nothing [S59]. Ask all hundred, again and again: *polling*.
3. **Let the kernel wait.** `select()` and `poll()` take a list of descriptors and sleep until one is "ready", meaning a read "without blocking" is possible [S47]. One sleeping call instead of thousands of asks. `epoll` is Linux's faster form of the same idea [S46].

# Predict: a hundred pipes, and nobody writing

Two programs, two seconds, zero bytes of data. They differ only in how they wait [S14].

## The setup

A hundred pipes. Nobody ever writes to them. Both programs wait two seconds, then exit.

## Program A

Asks every pipe in turn, non-blocking, round after round.

## Program B

Registers all hundred once, then makes one call that sleeps.

## The question

How many system calls does each one make? *About a hundred · a thousand · thirty thousand · millions.*

Two numbers, on paper. Neither program transfers a single byte.

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/>

# Where the question comes from

The cost of waiting, not of I/O.

|              |       |                                                       |          |
|--------------|-------|-------------------------------------------------------|----------|
| KEGEL, 1999  | [S55] | "So hardware is no longer the bottleneck."            | Named    |
| GAMMO ET AL. | [S49] | Plus 10,000 idle connections: "degrade by up to 79%". | Measured |

Up to 79% of select/poll throughput gone, only idle connections added [V61]. No answer made a read faster: each changed "**where the interest set lives**" [V62].

Kegel's 1999 revision lacks the idleness clause everyone quotes [S55] [V63].

[S55] The C10K problem, 8 May 1999 — Kegel. <https://web.archive.org/web/19990508164301/http://www.kegel.com/c10k.html> [S49] Gammo et al. <https://www.kernel.org/doc/ols/2004/ols2004v1-pages-215-226.pdf>

# A thread each, and why that stopped working

A hundred pipes, a hundred blocking threads — *thread-per-connection* — failed on two bills [S59].

## 1 Memory

Kegel priced the stacks: 2 MB each, "not an uncommon default value", against 1 GB of user-accessible address space. 512 threads, then no virtual memory left [V94].

## 2 Scheduler

Banga, Mogul and Druschel priced the rest: it "scales badly as the number of threads increases" [V95].

Either way the bill is per connection, not per byte [V96]. My note calls that "the same mistake at two altitudes" — my reading, not either source's [V96].

# Observe: two seconds, zero bytes, counted

|                  |       |        |         |               |       |        |             |
|------------------|-------|--------|---------|---------------|-------|--------|-------------|
| wait_blocking.py |       |        |         | wait_epoll.py |       |        |             |
| % time           | calls | errors | syscall | % time        | calls | errors | syscall     |
| 97.75            | 28017 | 28000  | read    | 16.97         | 100   |        | epoll_ctl   |
|                  |       |        |         | 1.97          | 1     |        | epoll_pwait |

## 28,017

read calls, 28,000 of them failing  
with EAGAIN [S53]

## 1

epoll\_pwait call, which slept for the  
whole two seconds [S53]

EAGAIN is not a fault. It is the kernel answering "nothing yet" to a non-blocking read, and `strace -c files` it under errors [S53].

[S53] demo/waiting/README.md — captured 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/waiting/README.md>

# What those counts do and do not say

Three things about 28,017 and 1 before you carry them anywhere [S53].

1. **28,017 grows with time.** Twenty seconds of polling makes ten times the calls, and every one of them learns nothing [S53].
2. **1 does not.** The `epoll` program paid a hundred registrations once, then one wait, however long it waits [S53].
3. **28,017 is too low.** The tracer slows every call. Untraced, the same loop "asked 1.3 million times in one second" [S53].

So: count under the tracer, time without it [S1]. Brendan Gregg measured a loop of nothing but system calls running 442× slower under `strace` [S66] [V17].

## epoll is three calls

1. `epoll_create1` makes an **interest set** that lives in the kernel between calls [S46].
2. `epoll_ctl` adds or removes one descriptor. Our hundred registrations [S53].
3. `epoll_wait` sleeps until a registered descriptor is readable. Our one call [S53].

`select` and `poll` hand the kernel the whole list on every call [S46]. `epoll` states it once, which is why Program B's count was a constant.

*Readable* has a precise meaning: a read "without blocking" is possible, because at least one byte is waiting or the writer has closed its end [S47]. It does not say how many bytes.

# Level-triggered and edge-triggered

One flag. Get it wrong and a connection goes quiet.

## LEVEL-TRIGGERED — THE DEFAULT

Unread data is reported on every wait. The manual: "a faster `poll(2)`" [S46].

## EDGE-TRIGGERED — EPOLLET

Reports only when something *new* arrives [S46]. Read 1 kB of a 2 kB arrival and the next wait "will probably hang" [S46].

**The rule for EPOLLET: every time you are woken, read everything — until EAGAIN — because the kernel will not mention the rest again [S46].** Non-blocking descriptors, or that loop never ends [S46].

# Three ways to wait, and we watch all three

One ladder. Each rung answers "who gets parked, and where" [V69].

- |          |                   |                                                                                   |
|----------|-------------------|-----------------------------------------------------------------------------------|
| <b>1</b> | <b>Polling</b>    | The program asks each descriptor. Our 28,000 failed reads [S53].                  |
| <b>2</b> | <b>Readiness</b>  | The kernel says "you may read now" [S46]. Our one <code>epoll_pwait</code> [S53]. |
| <b>3</b> | <b>Completion</b> | The program hands over a buffer. The kernel says "I have read for you" [V74].     |

In a trace: many failed reads means **polling**, one sleeping call means **readiness**, no reads at all during the wait means **completion**.

[S46] `epoll(7)` — Linux man-pages. <https://man7.org/linux/man-pages/man7/epoll.7.html>

# Readiness is permission to try, not a promise

The word *ready* sounds like a guarantee. It is not one.

`select(2)`, BUGS: it "may report a socket file descriptor as "ready for reading", while nevertheless a subsequent read blocks" [S47].

## 1 So use non-blocking mode

It is "safer to use `O_NONBLOCK` on sockets when blocking is undesirable" [S47].

## 2 The model

"permission to try, not a promise of success" [V58].

Ready reports a moment that has passed, on `select` and on `epoll` alike [V92]. A *done*, one rung up.

[S47] `select(2)` — Linux man-pages. <https://man7.org/linux/man-pages/man2/select.2.html>

# The real difference is who owns the buffer

Not the API shape. The ownership, and it is why porting between the models is a rewrite.

|                            | READINESS                                       | COMPLETION                                       |
|----------------------------|-------------------------------------------------|--------------------------------------------------|
| when you supply the buffer | after you learn the descriptor is ready [V75]   | at submission, before any data exists [V74]      |
| who owns it while waiting  | you — the kernel holds no pointer [V75]         | the kernel — you must not touch or free it [V74] |
| one buffer can             | be shared across ten thousand connections [V75] | serve one outstanding operation [V74]            |

My note calls ownership "the load-bearing row" [V57]. Microsoft's contract: clients "agree not to disturb the buffers in any way until the network has consumed the buffer's contents" [V74].

# It is not a Linux/Windows split

I have used the platforms as labels. Here is the correction.

- 1 The honest statement** "the readiness/completion split is not a Linux/Windows split" [V59].
- 2 The control case** BSD and macOS use kqueue, which is readiness [V81]. Not a Linux quirk [V59].
- 3 And both have crossed** Linux built a completion interface, Windows built a shared-memory ring [V68].

The useful question is not which system a program runs on. It is [which done it is handed](#).

# The trap this demonstration sets

We measured pipes. A regular file behaves differently, and the difference is silent.

- **Readiness and regular files**

[V73]

`epoll` "has nothing useful to say about a regular file, which is always "ready"". The call returns at once and you learn nothing.

- **Why a completion model exists at all**

[V88]

Linux "grew a completion interface because readiness could not express what its users needed".

So when you do meet these interfaces, do not expect a disk file in an `epoll` set to wait. It will not.

# Four Windows words before the Windows pair

The same two programs, in Windows vocabulary [S53].

1. **Named pipe.** "a named, one-way or duplex pipe for communication between the pipe server and one or more pipe clients" [S72]. A pipe with a name, openable by unrelated processes.
2. **PeekNamedPipe.** Looks without reading: it copies data "without removing it from the pipe" and "always returns immediately" [S73].
3. **Overlapped ReadFile.** A read that returns before the data exists; the kernel fills your buffer later [V74].
4. **Completion port, GetQueuedCompletionStatus.** A queue of finished operations; the call takes one, and "If there is no completion packet queued, the function waits" [S74].

# Predict: the same hundred pipes, on Windows

Completion is the rung we have not yet watched run [S14].

## Program C

A hundred named pipes. PeekNamedPipe on each, round after round — the Windows polling twin.

## Program D

One overlapped ReadFile posted per pipe, all bound to one completion port, then one GetQueuedCompletionStatus that sleeps.

## The question

Under NtTrace, how many native calls does each make in two seconds — and how many *reads* does Program D do while it waits?

Two numbers again. Both programs are in `demo/waiting/` [S53].

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/> [S53] `demo/waiting/README.md`, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/waiting/README.md>

# Observe: Windows, traced, both ways

|               |                   |               |                                       |
|---------------|-------------------|---------------|---------------------------------------|
| wait_peek.exe | 8153 native calls | wait_iocp.exe | 956 native calls                      |
| 2600          | NtFsControlFile   | 100           | NtReadFile (submitted up front)       |
| 2500          | NtCreateEvent     | 100           | NtSetInformationFile                  |
| 2725          | NtClose           | 1             | NtCreateIoCompletion                  |
|               |                   | 1             | NtRemoveIoCompletion ⇒ STATUS_TIMEOUT |

## 3

native calls per PeekNamedPipe:  
create an event, control the pipe,  
close [S53]

## 1

NtRemoveIoCompletion, which slept  
the whole two seconds [S53]

Untraced, the polling program made **330,100** peeks in two seconds. Traced, it made 2,500 [S53]. Count traced, time untraced.

[S53] demo/waiting/README.md — NtTrace64 captures, Windows 11 build 26200, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/waiting/README.md>

# Which numbers you may compare, and which you may not

Two counts per platform. Compare within a platform, never across [S53].

**You may, and shall, compare within one platform, both runs traced:**

- Linux: 28,017 read calls against 1 `epoll_pwait` [S53].
- Windows: 8,153 native calls, three per peek, against 956, of which one `NtRemoveIoCompletion` slept the whole two seconds [S53].

**You may not compare:** 28,017 with 330,100, since one is traced and one is not [S53]; or Linux with Windows, since they are different machines and read is not `PeekNamedPipe` [S53].

What survives: the same shape, twice. Thousands of asks against one sleeping call, on two kernels that agree on little else.

# What a ring is

One word before the last slide of Part 7.

A ring is two queues in memory that your program and the kernel both see: "ring buffers which are shared between user space and kernel space" [S75]. "You place I/O requests you want to make on the SQ, while the kernel places the results of those operations on the CQ" [S75].

Why it matters here: every wait we counted today was a system call. With a ring you queue many requests and cross the boundary once [S75], or, with a kernel-side polling thread, "steady-state I/O submission needs **no syscalls at all**" [V11].

# Where both platforms ended up

Not convergence. Each one took the half it was missing.

## LINUX

Readiness-native. Built a completion interface, `io_uring`, in kernel 5.1 [V79].

## WINDOWS

Completion-native since NT [V80]. Built a ring, `IoRing`, in Windows 11 build 22000 [V80].

"Linux adopted completion semantics; Windows adopted the shared-memory transport" [V68]. The ring and the completion model are two separable ideas, and each platform took the one it lacked.

Learn IOCP, the I/O completion port, as current Windows practice. Rings are "the direction of travel", not the exam [V60].

# Same mechanism, different clothes

---

A file, a pipe and a socket. One program, one trace, three destinations.

# Predict: which of the four calls changes?

You know the file run by heart now. I change only the destination [S14].

```
openat(AT_FDCWD, "no-such-file.txt", O_RDONLY) = -1 ENOENT
openat(AT_FDCWD, "iodemo.out", O_WRONLY|O_CREAT|O_TRUNC, 0666) = 3
write(3, "one\ntwo\nthree\n", 14)      = 14
close(3)                                = 0
```

I now run `./iodemo /dev/stdout | cat`. The bytes go into a pipe and cat reads them.

**On paper**

Which of the four calls changes — the `openat`, the `write`, the `close`, or nothing?

**And**

Does the byte count in the `write` change?

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/>

# Observe: the file run and the pipe run

```
= to a file =
openat(... "no-such-file.txt" ...) = -1
openat(... "iodemo.out" ...)      = 3
write(3, "one\ntwo\nthree\n", 14) = 14
close(3)                          = 0

= to a pipe =
openat(... "no-such-file.txt" ...) = -1
openat(... "/dev/stdout" ...)      = 3
write(3, "one\ntwo\nthree\n", 14) = 14
close(3)                          = 0
```

The same four calls, the same fourteen bytes in one write, the same descriptor number [S54]. cat on the other end received one two three.

**The `openat` argument changed.** The call did not, the count did not, and the byte total did not [S54].

[S54] demo/iodemo-c/README.md — the pipe run, captured 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/iodemo-c/README.md>

# The descriptor is the abstraction, not the file

The reason is a kernel structure, not a convention.

"On Linux, anything that takes a file descriptor takes a socket, because the kernel routes both through the same dispatch table" [V64]. A pipe too [V29].

So `write(3, ...)` said *done* when the bytes were in the pipe. Had `cat` read them yet? Probably not — but I traced only the writer, so that is my reading.

## 1

`write` call, for a file and a pipe alike  
[S54]

[S54] `demo/iodemo-c/README.md`, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/iodemo-c/README.md>

# Predict: the same program, writing to your terminal

`./iodemo /dev/stdout` with nothing after it. No pipe, no file. The screen [S14].

The same three `fwrite` calls: `"one\n"`, `"two\n"`, `"three\n"`. The file run and the pipe run each made **one** `write()` of fourteen bytes [S54].

**On paper: how many `write()` calls when the destination is a terminal — one, or three?** And a second answer: who decides, the kernel or the C library?

# Observe: three writes, not one

```
openat(AT_FDCWD, "/dev/stdout", O_WRONLY|O_CREAT|O_TRUNC, 0666) = 3
write(3, "one\n", 4)           = 4
write(3, "two\n", 4)           = 4
write(3, "three\n", 6)         = 6
close(3)                       = 0
```

Same binary, same container, run with a terminal attached (podman run -t); captured 2026-09-21 [S54]. Three system calls where the pipe run made one. The program did not change. The C library's decision did.

# Explain: the destination changed the trace, at the C library, not the kernel

"Attached to a terminal, stdout is line-buffered... Attached to a pipe or a regular file, stdout is fully (block) buffered" [V71].

1. **The library asks once**, at the first write, what the descriptor is. A terminal gets line buffering: every `\n` becomes a `write()` [V71].
2. **A file and a pipe** sit on the same side of that rule, so my first two runs agreed *by construction*: I chose two destinations the library treats the same.
3. **The kernel's part never changed**: `openat`, `write`, `close`. Only how many times.

Try it yourself: `./iodemo /dev/stdout` in a terminal, then with `| cat` [S54].

# Predict: the network leg

Same fourteen bytes, over a TCP socket, with netcat on both ends [S14].

## The setup

`./iodemo /dev/stdout | nc 127.0.0.1 9000`, with `nc -l` listening.  
Both netcats traced.

## The question

The listener receives the fourteen bytes from a TCP socket. Which call takes them off the wire?

## Two options

One of two. Write it down.

`read`, the same call you would use for a file — or a socket-specific call.

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/>

# Observe: both netcats, traced

```

= client nc (writer) =
socket(AF_INET, SOCK_STREAM...) = 3
connect(3, ...9000...) = -1 EINPROGRESS
read(0, "one\ntwo\nthree\n", 16384) = 14
write(3, "one\ntwo\nthree\n", 14) = 14
close(3) = 0

= listener nc (reader) =
socket(AF_INET, SOCK_STREAM...) = 3
bind(3, ...9000...) = 0
listen(3, 1) = 0
accept4(3, ...) = 4
read(4, "one\ntwo\nthree\n", ...) = 14

```

The client puts them on the wire with `write(3, ...)`. The listener takes them off with `read(4, ...)`. The same two calls `iodemo` used for a file and for a pipe [S54].

What is new is only how the descriptor came to exist [S54].

[S54] [demo/iodemo-c/README.md](https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/iodemo-c/README.md) — the traced netcat pair, captured 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/iodemo-c/README.md>

# Then what are send and recv for?

They are the program's choice of spelling, not the kernel's.

send(2) settles it in its own opening: "The only difference between send() and write() is the presence of flags. With a zero flags argument, send() is equivalent to write(2)" [S50].

- 1 Use them for a flag** MSG\_DONTWAIT and MSG\_NOSIGNAL have no write() equivalent.
- 2 Otherwise it is style** They tell a human reader "this is a socket".

[S50] send(2) — Linux man-pages. <https://man7.org/linux/man-pages/man2/send.2.html>

# The same interface is not the same contract

Three boundaries on the result.

- **One send is not one receive** [V70]  
TCP is a byte stream. Marking where a message ends is your job — "the single most common practical bug in student socket code".
- **A size effect** [V82]  
"writes of less than PIPE\_BUF bytes must be atomic" — 4 KiB here [V67] [V82].
- **The default is not total** [V87]  
"a strong default, not a total function" [V65]. `lseek` on a socket means nothing.

## Predict: the same three destinations, on Windows

On Linux the destination changed nothing in the trace. Windows is the contrast [S57].

- |   |                       |                                                  |
|---|-----------------------|--------------------------------------------------|
| 1 | Pipe                  | WriteFile to an anonymous pipe. Fourteen bytes.  |
| 2 | Socket                | send() on a loopback TCP socket, fourteen bytes. |
| 3 | The same socket again | WriteFile on that socket, cast to a HANDLE.      |

On paper: **is Windows the same as Linux here: only the destination changes, the call stays the same?**

**Yes or no, and if no, which leg differs?** One program, three destinations [S57].

[S57] demo/destinations-win32/README.md — captured 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/destinations-win32/README.md>

# Observe: the API changes the call, the object does not

```
NtWriteFile(FileHandle=0x118, Length=0xe, IoStatusBlock [0/0xe]) ⇒ 0
NtDeviceIoControlFile(FileHandle=0x138, IoControlCode=0x0001201f, ...) ⇒ 0
NtWriteFile(FileHandle=0x138, Length=0xe) ⇒ 0
```

|               |       |                                                |                |
|---------------|-------|------------------------------------------------|----------------|
| 1 · PIPE      | [S57] | NtWriteFile, 14 bytes                          | file path      |
| 2 · SEND()    | [S57] | NtDeviceIoControlFile,<br>control code 0x1201f | device control |
| 3 · WRITEFILE | [S57] | NtWriteFile — same<br>handle as leg 2          | file path      |

Legs 2 and 3 are one handle, two native calls. A plain WriteFile first failed with error 87 — Winsock sockets are **overlapped handles** [S57].

[S57] demo/destinations-win32/README.md, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/destinations-win32/README.md>

# Windows: a SOCKET is not a HANDLE

Third time today, and the same asymmetry each time.

## LINUX

A socket is a file descriptor, and Linux makes the file-descriptor contract binding [V28].

## WINDOWS

An opaque SOCKET, and Windows makes Winsock the contract [V28].

The NT kernel does hold a file object underneath [V28] — which is exactly what leg 3 just wrote to [S57]. The *type* is not a C-runtime file descriptor, while the *handle* is a kernel file handle [V66].

So the platforms agree on the mechanism and disagree on which layer they promise. Third time today.

# Bytes you never touch

---

One call can carry several buffers. And the kernel can move a file your process never holds.

# Predict: three buffers, one message

A message that is three pieces in your program, and one message in the file [S14].

## The program

A header, a body and a trailer — three separate buffers. You want all three in the file, in order, as one message.

## Question one

How many system calls does that take — *three, one, or zero*?

## Question two

How many times does anything copy the three buffers to make it happen?

Two answers, written separately. They are not the same question [S56].

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/> [S56] demo/scatter/README.md — captured 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/scatter/README.md>

# Observe: three lines become one

```
= three write() calls
write(3, "HEADER  ", 8)                = 8
write(3, "body body body body body body bo"... , 40) = 40
write(3, " TRAILER\n", 9)              = 9

= one writev() call (wrapped to fit)
writev(3, [{iov_base="HEADER  ", iov_len=8},
           {iov_base="body body body body body body bo"... , iov_len=40},
           {iov_base=" TRAILER\n", iov_len=9}], 3) = 57
```

Three crossings became **one**, and the return value is the whole message: 57 bytes [S56]. The three buffers never moved in the program.

[S56] demo/scatter/README.md — strace -e trace=write,writev, 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/scatter/README.md>

# Gather buys crossings, not copies

The distinction the name hides, and your second answer.

## WHAT WRITEV REMOVES

Two of the three user-to-kernel crossings, plus the staging buffer you would glue the pieces into [S58].

## WHAT IT DOES NOT REMOVE

Any copy the kernel makes further down. The four levels underneath are unchanged [S56].

`readv` is the mirror: one crossing *scatters* into several buffers [S58]. Valkey's reply path gathers framing, a stored string and a terminator into three `iovec` entries, then sends them with one `writenv` [S65].

[S58] `writenv(2)` — Linux man-pages. <https://man7.org/linux/man-pages/man2/writenv.2.html> [S56] [demo/scatter/README.md](https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/scatter/README.md), 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/scatter/README.md> [S65] [valkey/src/networking.c](https://github.com/valkey-io/valkey/blob/00a8a19554a39facb591837ea41ba4350260919d/src/networking.c) — Valkey (2026). <https://github.com/valkey-io/valkey/blob/00a8a19554a39facb591837ea41ba4350260919d/src/networking.c>

# Two promises, and only one of them is unconditional

*Atomic* means one thing: your bytes land as one unbroken block, with nobody else's write in the middle [S58].

1. **On a regular file** the promise is unconditional: "written as a single block that is not intermingled with output from writes in other processes" [S58]. A log line written with one `writew` stays whole.
2. **On a pipe** it holds only below `PIPE_BUF`, 4096 bytes. Above that the kernel "may interleave the data" with another writer's [S59].
3. **The trap:** the limit applies to the *sum* of your buffers, not the largest. Eight 1 KiB pieces are 8 KiB, and may interleave.

# Predict: moving a file you never look at

4 MiB from a file into a pipe. Two programs, same bytes [S14].

## Program A

A copy loop: read 64 KiB into a buffer, write it out, repeat.

## Program B

One call per chunk: `sendfile(out, in, ...)`. The program never allocates a buffer for the data.

## The question

How many data system calls does each make — and how many times does each byte cross into your address space?

Neither program looks at a single byte of the file [S56].

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/> [S56] demo/scatter/README.md — captured 2026-09-20. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/scatter/README.md>

# Observe: half the calls, and no trip through your program

| 4 MIB INTO A PIPE        | DATA SYSCALLS                   | COPIES THROUGH USER SPACE | WALL   |
|--------------------------|---------------------------------|---------------------------|--------|
| copy loop, 64 KiB chunks | 64 read + 64 write = <b>128</b> | two per chunk             | 2.6 ms |
| sendfile                 | <b>64</b> sendfile              | none                      | 2.0 ms |

The trace's tell is an absence: with `sendfile` there is **no read line at all** [S56]. The bytes went from the page cache to the pipe without visiting your address space.

Sixty-four calls because each moved 64 KiB — the pipe's capacity, not a `sendfile` limit [S56] [S59].

[S56] [demo/scatter/README.md](https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/scatter/README.md) — `strace -c -e trace=read,write,sendfile, 2026-09-20`. <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/scatter/README.md> [S59] `pipe(7)` — Linux man-pages. <https://man7.org/linux/man-pages/man7/pipe.7.html>

# What the name does not cover

It counts the copies it removes at one boundary. No others.

- |          |                                   |                                                                         |
|----------|-----------------------------------|-------------------------------------------------------------------------|
| <b>1</b> | <b>Direct memory access stays</b> | The bytes still move. The CPU no longer carries them [S60].             |
| <b>2</b> | <b>It needs the page cache</b>    | Level 3 is not bypassed. Handing out its pages <i>is</i> the mechanism. |
| <b>3</b> | <b>It buys no durability</b>      | The same four levels sit below [S67].                                   |
| <b>4</b> | <b>It cannot help you look</b>    | Encrypt them and the process must hold them [S60].                      |

[S60] sendfile(2). <https://man7.org/linux/man-pages/man2/sendfile.2.html> [S67] fsync(2). <https://man7.org/linux/man-pages/man2/fsync.2.html>

# Zero-copy moves bytes faster. It does not make them safer.

Fewer copies changes *who* copies, not *when the bytes are safe*.

1. **To a file.** `sendfile` copies "within the kernel" [S60], into the page cache. Nothing is on the device until `fsync()` [S67].
2. **To a socket.** The return counts bytes the kernel accepted, not delivered: "No indication of failure to deliver is implicit in a `send()`" [S50].
3. **One new duty.** The kernel may still be reading *your* pages after the call returns: keep the file "unmodified until the reader on the other end of `out_fd` has consumed the transferred data" [S60].

*Done* still means *accepted* [V44].

# Windows, and a silent fallback

Other names, and a Python fallback that says nothing.

- 1**            **Gather, a socket**            `WSASend` with an array of `WSABUF` [S69].
- 2**            **Gather, a file**            `WriteFileGather`, but every buffer must be at least an aligned page [S62].
- 3**            **Zero-copy**            `TransmitFile` [S63].

Without `os.sendfile()`, `socket.sendfile()` uses "send() ... as a less efficient fallback" [S64]. **No warning:** on Windows you measure that.

[S62] `WriteFileGather` — Microsoft Learn. <https://learn.microsoft.com/en-us/windows/win32/api/fileapi/nf-fileapi-writefilegather> [S63]  
`TransmitFile` — Microsoft Learn. <https://learn.microsoft.com/en-us/windows/win32/api/mswsock/nf-mswsock-transmitfile> [S64] `socket` — Python docs. <https://docs.python.org/3/library/socket.html#socket.socket.sendfile>

# Homework, and the next question

---

What changed about the assignment, what you can run this week, and where Lecture 4 picks up.

# Assignment 1 is not set today

I told you 22 September in Lecture 1. That has moved, and here is the honest reason.

## What changed

The submission infrastructure is not ready. The assignments move back.

## New dates

None yet. I will not put a date on a slide before I can keep it.

## What it will measure

Buffered against unbuffered, and sequential against random access, over a 1 MiB file. Predict the ratio. Measure it. Explain it in crossings.

## Same method

Predict, observe, explain [S14].

[S14] Predict-Observe-Explain — Taber. <https://science-education-research.com/teaching-science/constructivist-pedagogy/predict-observe-explain/>

# Voluntary homework, and three rules for it

Everything I ran today ships with this deck. Run it yourself.

- 1 Predict, then look** Write the number first. Watching alone teaches nothing [S51].
- 2 Report version and buffer size** Python's default binary buffer moved from 8 KiB to 128 KiB [S7].
- 3 Count traced, time untraced** Two runs. One for counts, one for timings [S1].  
`demo/durability/ · demo/waiting/ · demo/iodeemo-c/ · demo/scatter/` run in the dev container.  
`demo/destinations-win32/` needs Windows [S57].

[S51] Crouch, Fagen, Callan and Mazur (2004). <https://works.swarthmore.edu/fac-physics/203/> [S7] CPython gh-117151. <https://github.com/python/cpython/issues/117151> [S1] strace(1). <https://man7.org/linux/man-pages/man1/strace.1.html>

# Summary — what follows from this

- 1**      **A return value is a handoff**      It names the layer that took them [V50].
- 2**      **Name the boundary first**      "I flushed" is ambiguous across four [V44].
- 3**      **Design for a wrong *done***      Three engines did, in a year [V40] [V56].
- 4**      **Crossings and copies differ**      `writenv` removed crossings, `sendfile` copies [S56].  
Every number today came from a run [S52] [S53] [S56]. Whose *done* did you measure? [V50]

# What Lecture 4 answers

Same pattern as last week. I name it and do not answer it.

## Today's window

Your bytes sat in the page cache for five seconds, in memory no process owns [S52].

## In part 9

The kernel moved four mebibytes you never held — the same pages [S56].

## The question

What *is* that memory, and whose is it?

## Why it is not obvious

"Physical memory is a cache, not a container" [V72].

If we did not reach parts 7 to 9, Lecture 4 opens with them.

[S52] [demo/durability/README.md](https://demo/durability/README.md). <https://ondrasek.github.io/talks/2026-09-22-pb158-lecture-3-finishing-io-and-advanced-io/demo/durability/README.md>

# References · 1/27

- **[S1]** `strace(1)` — Linux man-pages — <https://man7.org/linux/man-pages/man1/strace.1.html>
- **[S2]** `write(2)` — Linux man-pages — <https://man7.org/linux/man-pages/man2/write.2.html>
- **[S3]** `fflush(3)` — Linux man-pages — <https://man7.org/linux/man-pages/man3/fflush.3.html>
- **[S4]** `close(2)` — Linux man-pages — <https://man7.org/linux/man-pages/man2/close.2.html>
- **[S5]** C99 draft, §7.19.3 — <https://busybox.net/~landley/c99-draft.html>
- **[S6]** Built-in Functions — `open()`, Python documentation — <https://docs.python.org/3/library/functions.html#open>
- **[S7]** Increase `io.DEFAULT_BUFFER_SIZE` — CPython issue 117151 — <https://github.com/python/cpython/issues/117151>

## References · 2/27

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- **[S11]** End-to-End Arguments in System Design — <https://web.mit.edu/Saltzer/www/publications/endtoend/endtoend.pdf>
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- **[S32]** ltrace — project page — <https://ltrace.org/>
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- **[S71]** PostgreSQL commit 9ccdd7f — "PANIC on fsync() failure" (2018) — <https://git.postgresql.org/gitweb/?p=postgresql.git;a=commit;h=9ccdd7f66e3324d2b6d3dec282cfa9ff084083f1>
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- **[S74]** GetQueuedCompletionStatus function (ioapiset.h) — Microsoft Learn — <https://learn.microsoft.com/en-us/windows/win32/api/ioapiset/nf-ioapiset-getqueuedcompletionstatus>
- **[S75]** io\_uring(7) — Linux man-pages — [https://man7.org/linux/man-pages/man7/io\\_uring.7.html](https://man7.org/linux/man-pages/man7/io_uring.7.html)
- **[V4]** A Successful write() Is a Handoff, Not a Save — The idiom the room already writes contains no durability step.
- **[V5]** A Successful write() Is a Handoff, Not a Save — The limit on the thesis. The bottom of the stack is not observable from software, and the deck says so.

## References · 13/27

- **[V6]** Standard output is buffered differently depending on where it points — Removes the misconception that a Python write passes through the C library buffer.
- **[V8]** Windows and Linux place the stable syscall boundary at different layers — Takeaway 4, stated as a placement question rather than a quality judgement.
- **[V9]** Windows and Linux place the stable syscall boundary at different layers — Stops section 5 reading as a complaint about Windows.
- **[V10]** The System Call Boundary — Makes "a library call is not a system call" falsifiable with two tools instead of asserting it.
- **[V11]** The System Call Boundary — The closing honest-limits slide.

## References · 14/27

- **[V14]** Where the Abstraction Leaks — Bounds section 3's vanishing-output demonstration so it does not teach the misconception it exists to remove.
- **[V17]** Observing a Running System — Why sections 1 and 2 need two runs.
- **[V20]** File IO and Durability — The device reports success before the bytes reach the medium, which is the sentence Lecture 2 stopped on.
- **[V21]** NtTrace — Native API tracing for Windows (Roger Orr) — Why this tool is usable on a student's own machine, and what its design bought.
- **[V22]** NtTrace — Native API tracing for Windows (Roger Orr) — The single string in the Windows trace that shows the layer difference.

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- **[V23]** NtTrace — Native API tracing for Windows (Roger Orr) — Comparison slide names the defect it has just cleared.
- **[V24]** PostgreSQL's fsync() surprise — Shipped storage engines answer the deferred-error problem by refusing to retry.
- **[V25]** Every Tracing Mechanism Trades Observation Cost for a Blind Spot — The honest-limits slide. The tool taught here is a teaching and debugging instrument, not an operational one.
- **[V28]** Sockets and Network I-O — One-slide bridge to networking.
- **[V29]** Sockets and Network I-O — The pipe row on the same slide.
- **[V30]** File IO and Durability — The row of the note's per-layer instrumentation table covering the kernel-to-device boundary.

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- **[V33]** File IO and Durability — The bottom arrow of the stack diagram, and the "emptied by" cell of level 4.
- **[V36]** Windows and Linux place the stable syscall boundary at different layers — The caveat on the outlier list.
- **[V37]** File IO and Durability — The Windows half of level 3's "emptied by" cell, and of the third arrow in the stack diagram.
- **[V38]** Windows and Linux place the stable syscall boundary at different layers — Why Linux nominates the layer it does, and why the other platforms need not.
- **[V39]** Every Tracing Mechanism Trades Observation Cost for a Blind Spot — What the lower-overhead mode named in [V25] actually buys, on a named workload.

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- **[V40]** PostgreSQL's fsync() surprise — How many storage engines abandoned the retry, and when.
- **[V41]** PostgreSQL's fsync() surprise — The independence of the three engines, which is what makes the never-retry slide an argument rather than an anecdote.
- **[V43]** Windows and Linux place the stable syscall boundary at different layers — The named list behind the outlier correction.
- **[V44]** File IO and Durability — Core claim, stated as a conjunction.
- **[V45]** File IO and Durability — The objection the room raises during section 3 — *but I can read the file back.*
- **[V46]** File IO and Durability — Windows half at equal depth.

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- **[V47]** File IO and Durability — The honest Windows answer is *undocumented*, which is a stronger and truer statement than *the same* or *safe*.
- **[V48]** File IO and Durability — Honesty slide. Why the 40 MiB file was written back sooner than the 4 MiB one, which is the first question the observation provokes.
- **[V49]** File IO and Durability — Close. A shipped database declines to trust the last software handoff and defends itself another way.
- **[V50]** Where the Abstraction Leaks — The thesis in one sentence, from the vault's own synthesis note.
- **[V51]** Where the Abstraction Leaks — Fixable half. The room needs the part software *can* get right, or the durability section reads as fatalism.

## References · 19/27

- **[V52]** Where the Abstraction Leaks — Counter-position. Trusting the abstraction is respectable, and the note says exactly where it stops being respectable.
- **[V53]** A failed fsync() is not retryable — The kernel's *done* observed lying, in a trace of exactly the kind this course teaches the room to read.
- **[V54]** A failed fsync() is not retryable — Why the obvious repair is not a repair. It has the same shape as Lecture 2's slide "Never retry a failed close", one level down.
- **[V55]** A failed fsync() is not retryable — Windows slide. The asymmetry is about documentation, not about which platform is safer.
- **[V56]** PostgreSQL's fsync() surprise — The industry answer to an untrustworthy *done* is crash and replay, and it was never meant to last this long.

## References · 20/27

- **[V57]** Readiness and Completion Models — Buffer ownership, not API taste, is what separates the two models, and it is why porting between them is a rewrite.
- **[V58]** Readiness and Completion Models — The sentence that stops the room reading *ready* as a guarantee, which is the misconception this model produces.
- **[V59]** Readiness and Completion Models — Correction to its own framing.
- **[V60]** Readiness and Completion Models — Windows slide. What the room should carry away as present tense, and what is only direction of travel.
- **[V61]** The C10K Problem Was About the Cost of Waiting, Not I-O — Predict slide. The controlled experiment behind the question, stated as what was held fixed and what was varied.

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- **[V62]** The C10K Problem Was About the Cost of Waiting, Not I-O — Explanation. epoll did not make a read faster, and the room needs that said before it draws the wrong conclusion from the counts.
- **[V63]** The C10K Problem Was About the Cost of Waiting, Not I-O — The page most often cited for this argument did not make it in 1999, so the deck cites the revision and the measurement instead.
- **[V64]** Sockets and Network I-O — Mechanism. Why read and write work on a socket, stated as a kernel structure rather than as a convention.
- **[V65]** Sockets and Network I-O — Boundary. The same interface does not mean the same contract, and the note names where the default stops.
- **[V66]** Sockets and Network I-O — Windows slide. The platforms agree on the mechanism and disagree on which layer they promise, which is the accurate statement.

## References · 22/27

- **[V67]** Interprocess Communication — Boundary. The clean one-write-one-read result in the capture holds because fourteen bytes is small, and the room should know the threshold.
- **[V68]** io\_uring and Windows IoRing — Completion on Both Platforms — Close. Where both platforms ended up, stated precisely enough not to read as "they converged".
- **[V69]** Blocking Is a Scheduler Decision, Not a Property of the I-O — The one question that orders polling, readiness and completion into a ladder rather than a feature list.
- **[V70]** Network Buffering and Delivery Semantics — Boundary. The capture shows one write and one read only because the payload is fourteen bytes, and the slide must say so.
- **[V71]** Standard output is buffered differently depending on where it points — Honest bound on its own demonstration.

## References · 23/27

- **[V72]** Virtual Memory and the Page Cache — Handover. The page cache held the bytes for five seconds, and what that memory actually is belongs to the next lecture.
- **[V73]** io\_uring and Windows IoRing — Completion on Both Platforms — Boundary. The demonstration uses pipes, and a student who tries the same loop on a file gets a mechanism that always answers yes.
- **[V74]** Readiness and Completion Models — What a completion model demands of the program, in Microsoft's own words, and the memory bill that follows from it.
- **[V75]** Readiness and Completion Models — The other half of the ownership comparison.
- **[V76]** Readiness and Completion Models — Why the choice of model is structural rather than a matter of API taste.
- **[V77]** Readiness and Completion Models — Takeaway 2 and section 4's definition slide.

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- **[V78]** File IO and Durability — What a dirty page is, what writeback is, and that the kernel rather than the program schedules it.
- **[V79]** io\_uring and Windows IoRing — Completion on Both Platforms — Close. The version in which Linux acquired a completion interface.
- **[V80]** io\_uring and Windows IoRing — Completion on Both Platforms — Close. The build in which Windows acquired a ring, paired with [V79].
- **[V81]** Readiness and Completion Models — Correction. The control case that shows readiness is the Unix family's answer rather than a Linux quirk.
- **[V82]** Interprocess Communication — Boundary. The threshold in bytes, and what happens above it, which the atomicity span [V67] does not state.

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- **[V83]** A failed `fsync()` is not retryable — Windows slide. The mechanism behind the undocumented contract, which is why no single answer exists.
- **[V84]** PostgreSQL's `fsync()` surprise — What InnoDB actually changed, which the count-and-year span [V40] does not carry.
- **[V85]** A failed `fsync()` is not retryable — Close. The log-and-replay answer, in PostgreSQL's own shipped documentation rather than as the deck's summary of it.
- **[V86]** File IO and Durability — The device lying is a reported practice rather than a theoretical possibility, and a shipped database says so.
- **[V87]** Sockets and Network I-O — Boundary. A concrete operation that does not transfer, which the strong-default span [V65] asserts without instancing.

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- **[V88]** `io_uring` and Windows `IoRing` — Completion on Both Platforms — Why the regular-file gap mattered enough to build a second model, which the `epoll-and-regular-files` span [V73] does not say.
- **[V89]** A failed `fsync()` is not retryable — The mechanism behind the failed retry, which [V54]'s one-sentence span states without explaining.
- **[V90]** A failed `fsync()` is not retryable — How long the misreading stood before anyone noticed, which is what makes it a contract failure rather than a careless bug.
- **[V91]** File IO and Durability — Windows slide. The judgement itself, from the note that makes it, rather than asserted by the deck on the strength of a missing wiki row.
- **[V92]** Readiness and Completion Models — The `select(2)` BUGS caveat applies to the interface the demonstration actually used.

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- **[V93]** PostgreSQL's fsync() surprise — What PostgreSQL actually changed, and that the change reached every release rather than only the current one.
- **[V94]** The C10K Problem Was About the Cost of Waiting, Not I-O — Kegel's arithmetic for why a thread per connection ran out of virtual memory.
- **[V95]** The C10K Problem Was About the Cost of Waiting, Not I-O — Banga, Mogul and Druschel on the scheduler half of the same bill.
- **[V96]** The C10K Problem Was About the Cost of Waiting, Not I-O — What unifies the thread rung with the polling rung: a charge levied per connection, not per byte.

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